9 The table shows the probability distribution for the discrete random variable X.

x	1	2	3	4	5
$P(X = x)$	0.1	0.3	q	$2q$	$3q$

You are given that q is a positive constant.

(a) Determine the value of q. **[2]**

(b) Calculate $P(X \leqslant 4)$. **[1]**

Two independent values of X are taken.

(c) Determine the probability that the sum of the two values is 3. **[2]**

Fifty independent values of X are taken.

(d) Find the probability that a value of 2 occurs exactly 17 times. **[1]**

9(a)

9(b)

9(c)

9(d)